5.6 어떤 주기 신호의 삼각함수 형식 푸리에 급수가 다음과 같다.

$$x(t) = 3 + \sqrt{3}\cos(2t) + \sin(2t) + \sin(3t) - \frac{1}{2}\cos\left(5t + \frac{\pi}{3}\right)$$

(a) 간결형 삼각함수 형식 푸리에 스펙트럼을 그려라.
(b) (a)의 결과로부터 지수함수 형식 푸리에 스펙트럼을 그려라.
(c) (b)의 스펙트럼을 보고 $x(t)$의 지수함수 형식 푸리에 급수를 구하라.

(a) DC 성분: $A_0 = 3$

기본각주파수 $\omega_0 = 1$ (rad/s)

조화성분 $n = 2, 3, 5$ (음의 차수로 대칭을 준다)

$n=2$ 성분 합성

$$\sqrt{3}\cos(2t) + \sin(2t) = A_2\cos(2t + \theta_2)$$

$$\Downarrow \quad A_2\cos(2t+\theta_2) = A_2[\cos\theta_2\cos 2t - \sin\theta_2 \sin 2t]$$

$$\begin{cases} A_2\cos\theta_2 = \sqrt{3} \\ -A_2\sin\theta_2 = 1 \end{cases}$$

1. 진폭 A_2 계산

$$A_2 = \sqrt{\sqrt{3}^2 + 1^2} = 2$$

2. 위상 θ_2 계산

$$\tan\theta_2 = \frac{-1/A_2}{\sqrt{3}/A_2} = \frac{1}{\sqrt{3}} \Rightarrow \theta_2 = -\frac{\pi}{6}$$

따라서

$$\sqrt{3}\cos(2t) + \sin(2t) = 2\cos\left(2t - \frac{\pi}{6}\right)$$

$n=3$ 성분 합성

$\sin(3t)$를 코사인 단일 위상 합성

$$\sin(3t) = \cos\left(3t - \frac{\pi}{2}\right)$$

$$A_3 = 1, \quad \theta_3 = -\frac{\pi}{2}$$

$n=5$ 성분 합성

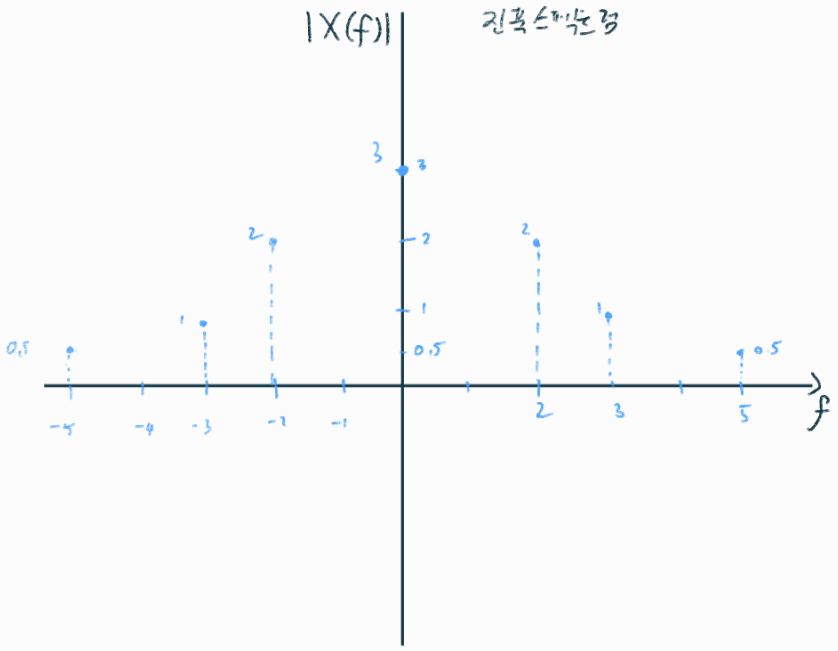
$$-\frac{1}{2}\cos\left(5t + \frac{\pi}{3}\right) = A_5\cos(5t + \theta_5)$$

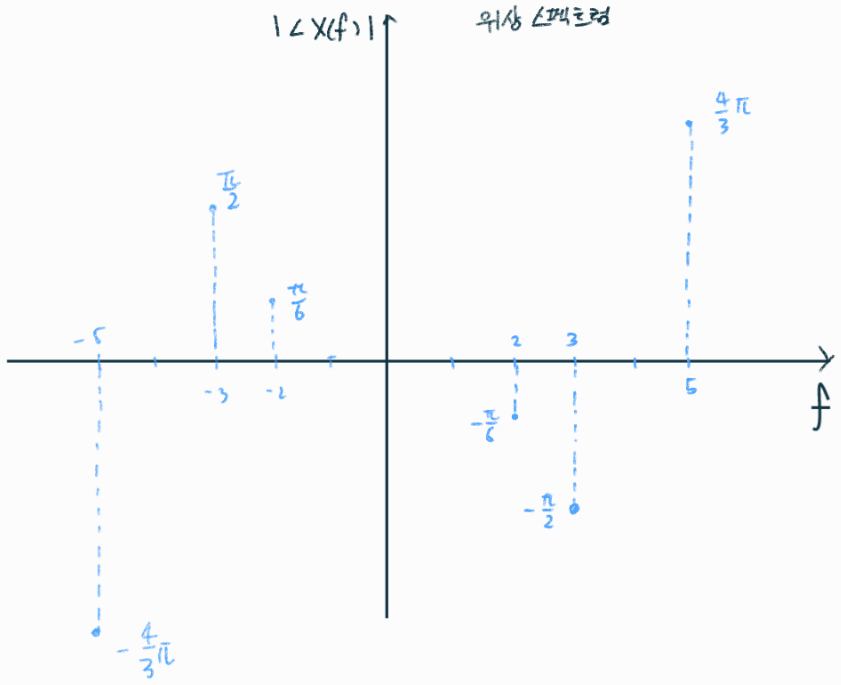
$$\rightarrow \quad -\frac{1}{2}\cos\left(5t + \frac{\pi}{3}\right) = \frac{1}{2}\cos\left(5t + \frac{\pi}{3} + \pi\right) = \frac{1}{2}\cos\left(5t + \frac{4}{3}\pi\right)$$

$$A_5 = \frac{1}{2}, \quad \theta_5 = \frac{4}{3}\pi$$

차수	진폭	위상
0	3	—
± 2	2	$\mp\frac{\pi}{6}$

± 3	1	$\mp\frac{\pi}{2}$
± 5	0.5	$\pm\frac{4\pi}{3}$





(b) DC 계수

$$C_0 = 3,\quad |C_0| = 3,\quad \angle C_0 = 0$$

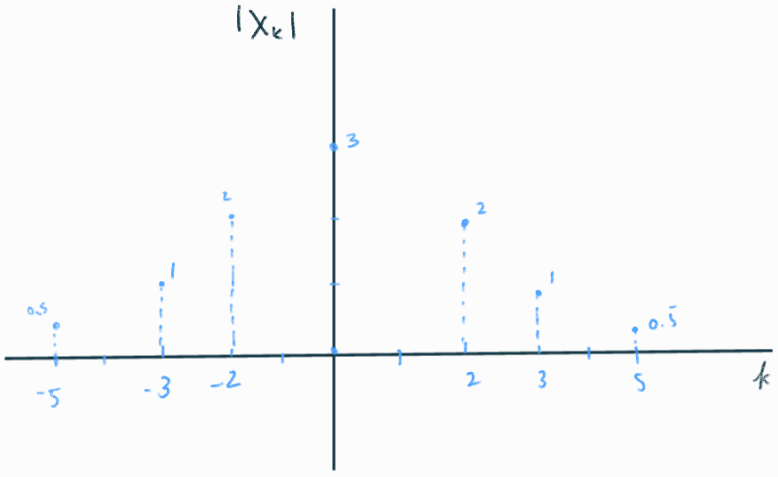
조화 성분 계수

$$C_{\pm 2} = \frac{A_2}{2} e^{\pm j\theta_2} = 1 \cdot e^{\mp j\pi/6},\quad |C_{\pm 2}| = 1,\quad \angle C_2 = -\frac{\pi}{6},\quad \angle C_{-2} = \frac{\pi}{6}$$

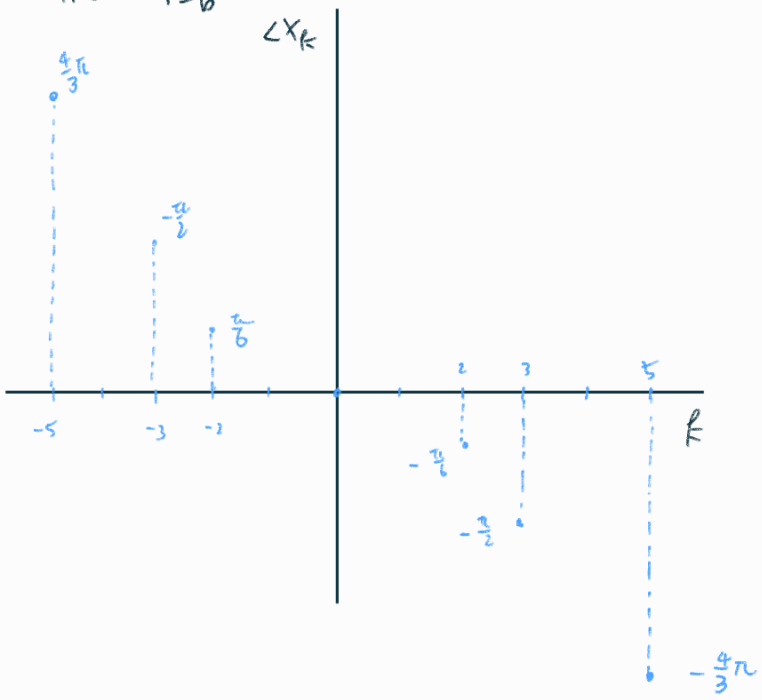
$$C_{\pm 3} = \frac{A_3}{2} e^{\pm j\theta_3} = \frac{1}{2} e^{\mp j\pi/2},\quad |C_{\pm 3}| = \frac{1}{2},\quad \angle C_3 = -\frac{\pi}{2},\quad \angle C_{-3} = +\frac{\pi}{2}$$

$$C_{\pm 5} = \frac{A_5}{2} e^{\pm j\theta_5} = \frac{1}{4} e^{\pm j(4\pi/3)}\quad |C_{\pm 5}| = \frac{1}{4},\quad \angle C_5 = +\frac{4\pi}{3},\quad \angle C_{-5} = -\frac{4\pi}{3}$$

◦ 진폭 스펙트럼



• 위상 스펙트럼



(1)

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}, \quad \omega_0 = 1$$

이고 나머지 $C_n = 0$ (즉 $n \neq 0, \pm 2, \pm 3, \pm 5$)

$C_0 = 3$

$C_{\pm 2} = \cos\frac{\pi}{6} \mp j\sin\frac{\pi}{6}$

$C_{\pm 3} = \mp\frac{j}{2}$

$C_{\pm 5} = -\frac{1}{8} \mp j\frac{\sqrt{3}}{8}$

$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$

$\therefore x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}, \quad \omega_0 = 1$

$= 3 + e^{-j\frac{\pi}{6}} e^{j2t} + e^{j\frac{\pi}{6}} e^{-j2t} + \frac{1}{2} e^{-j\frac{\pi}{2}} e^{-j3t}$

$+ \frac{1}{2} e^{j\frac{\pi}{2}} e^{-j3t} + \frac{1}{4} e^{j\frac{4}{3}\pi} e^{j5t} + \frac{1}{4} e^{-j\frac{4}{3}\pi} e^{-j5t}$

5.7 다음의 주기 신호에 대해 지수함수 형식 푸리에 급수를 구하고 스펙트럼을 그려라.

(a) $x(t) = 2\sin(\pi t) + \cos(3\pi t) - \cos(5\pi t)$

(b) $x(t) = 1 + 4\cos\left(\pi t - \frac{\pi}{4}\right) - \cos\left(3\pi t + \frac{\pi}{3}\right)$

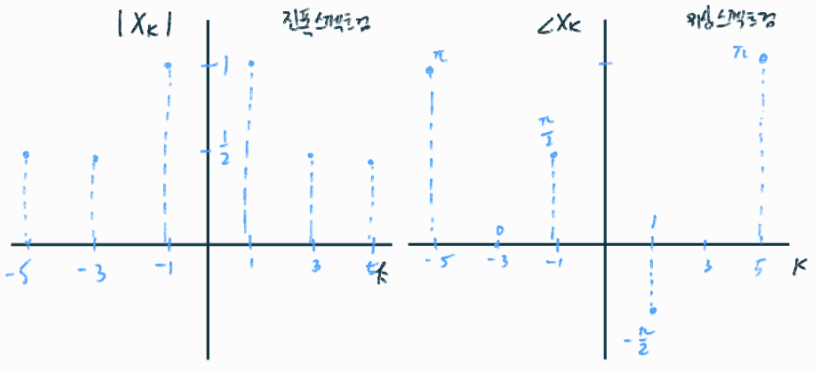
(c) $x(t) = \cos\left(2\pi t + \frac{\pi}{4}\right) - \sin(4\pi t) + \sin\left(6\pi t - \frac{\pi}{3}\right)$

(d) $x(t) = 1 + 4\sin\left(t - \frac{\pi}{4}\right) - 2\cos(2t) + \sin\left(3t + \frac{\pi}{3}\right)$

(a)

	위상
$2\sin(\pi t) = -je^{j\pi t} + je^{-j\pi t}$	$\pm\frac{\pi}{2}$
$\cos(3\pi t) = \frac{e^{j3\pi t} + e^{-j3\pi t}}{2}$	0
$-\cos(5\pi t) = -\frac{e^{j5\pi t} + e^{-j5\pi t}}{2}$	π

✓ $x(t) = -je^{j\pi t} + je^{-j\pi t} + \frac{e^{j3\pi t} + e^{-j3\pi t}}{2} - \frac{e^{j5\pi t} + e^{-j5\pi t}}{2}$



(b)

$\cos\left(2\pi t + \frac{\pi}{4}\right) = \frac{1}{2} e^{j\frac{\pi}{4}} e^{j2\pi t} + \frac{1}{2} e^{-j\frac{\pi}{4}} e^{-j2\pi t}$

$C_{+1} = \frac{1}{2} e^{j\frac{\pi}{4}} \qquad C_{-1} = \frac{1}{2} e^{-j\frac{\pi}{4}}$

위상 $= \frac{\pi}{4}$ 위상 $= -\frac{\pi}{4}$

$-\sin(4\pi t) = \frac{j}{2} e^{j4\pi t} - \frac{j}{2} e^{-j4\pi t}$

$C_{+2} = \frac{j}{2} \qquad C_{-2} = -\frac{j}{2}$

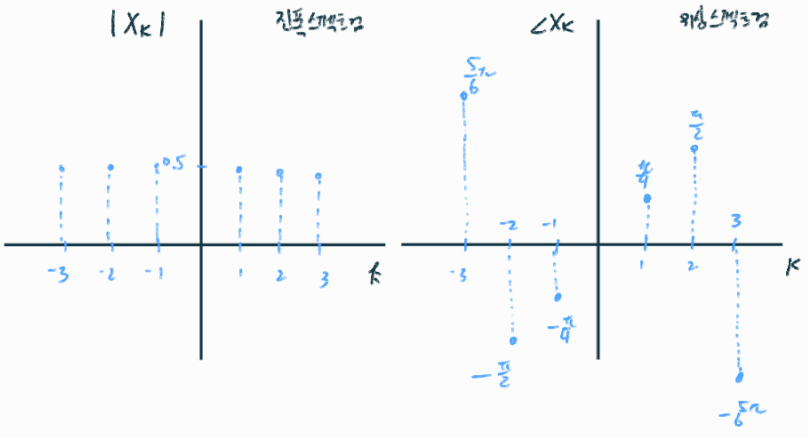
위상$_0 = -\frac{\pi}{2}$

$\sin(6\pi t - \frac{\pi}{3}) = -\frac{j}{2}e^{-j\frac{\pi}{3}}e^{j6\pi t} + \frac{j}{2}e^{j\frac{\pi}{3}}e^{-j6\pi t}$

$C_{+3} = -\frac{j}{2}e^{-j\frac{\pi}{3}}$ $\quad C_{-3} = \frac{j}{2}e^{j\frac{\pi}{3}}$

위상$_0 = -\frac{\pi}{2} - \frac{\pi}{3} = -\frac{5\pi}{6}$ $\quad$ 위상$_0 = \frac{5}{6}\pi$

$x(t) = \frac{1}{2}e^{j\frac{\pi}{4}}e^{j2\pi t} + \frac{1}{2}e^{-j\frac{\pi}{4}}e^{-j2\pi t} + \frac{j}{2}e^{j4\pi t} - \frac{j}{2}e^{-j4\pi t} - \frac{j}{2}e^{-j\frac{\pi}{3}}e^{j6\pi t}$
$+ \frac{j}{2}e^{j\frac{\pi}{3}}e^{-j6\pi t}$



5.9 주기 신호의 지수함수 형식 푸리에 급수가 다음과 같을 때, 실수 주기 신호를 구하라.

(a) $x(t) = e^{j\frac{\pi}{2}}e^{-j3\pi t} + 2e^{j\frac{\pi}{3}}e^{-j\pi t} + 3 + 2e^{-j\frac{\pi}{3}}e^{j\pi t} + e^{-j\frac{\pi}{2}}e^{j3\pi t}$

(b) $x(t) = -je^{-j4\pi t} + (1-j1)e^{-j2\pi t} + 2 + (1+j1)e^{j2\pi t} + je^{j4\pi t}$

(a)

C_n

$C_{-3} = e^{j\frac{\pi}{2}}$ $\quad C_{-1} = 2e^{j\frac{\pi}{3}}$ $\quad C_0 = 3$

$C_{+3} = e^{-j\frac{\pi}{2}}$ $\quad C_{+1} = 2e^{-j\frac{\pi}{3}}$ $\quad C_n = 0$ (otherwise)

$y(t) = \sum_{n=-3}^{3} C_n e^{jn\pi t} = C_0 + [C_{+1}e^{j\pi t} + C_{-1}e^{-j\pi t}]$
$+ [C_{+3}e^{j3\pi t} + C_{-3}e^{-j3\pi t}]$

$n = 1$

$C_{+1}e^{j\pi t} + C_{-1}e^{-j\pi t} = 2|C_1|\cos(\pi t + \arg C_1)$

$|C_1| = 2, \quad \arg C_1 = -\frac{\pi}{3}$

$$\therefore\ C_{+1}e^{j\pi t} + C_{-1}e^{-j\pi t} = 4\cos\left(\pi t - \frac{\pi}{3}\right)$$

$n=3$

$$C_{+3}e^{j3\pi t} + C_{-3}e^{-j3\pi t} = 2|C_3|\cos(3\pi t + \arg C_3)$$

$$|C_3| = 1,\ \arg C_3 = -\frac{\pi}{2}$$

$$\therefore\ C_{+3}e^{j3\pi t} + C_{-3}e^{-j3\pi t} = 2\cos\left(3\pi t - \frac{\pi}{2}\right)$$

$$\therefore\ x(t) = 3 + 4\cos\left(\pi t - \frac{\pi}{3}\right) + 2\cos\left(3\pi t - \frac{\pi}{2}\right)$$

(b) C_n

$C_{-2} = -j \quad C_{-1} = 1-j \quad C_0 = 2$

$C_{+2} = j \quad C_{+1} = 1+j \quad C_n = 0 \quad (\text{otherwise})$

$n=1$

$$C_{+1}e^{j\omega_0 t} + C_{-1}e^{-j\omega_0 t} = 2|1+j|\cos(2\pi t + \arg(1+j))$$

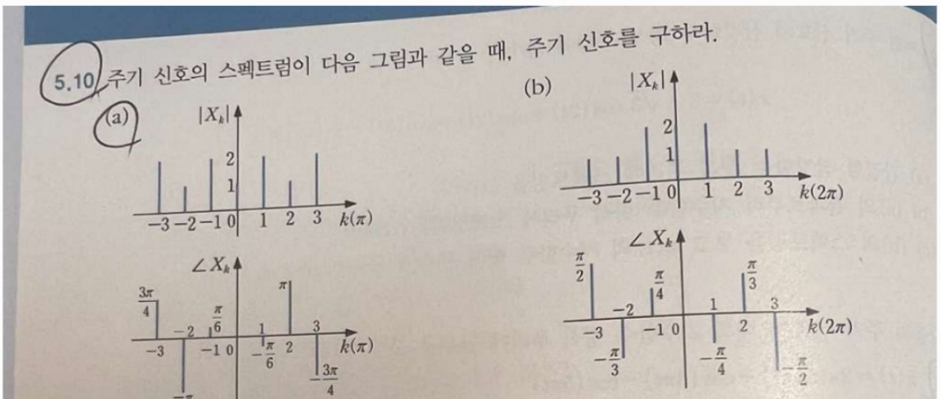
$$|1+j| = \sqrt{2},\ \arg(1+j) = \frac{\pi}{4} \rightarrow 2\sqrt{2}\cos\left(2\pi t + \frac{\pi}{4}\right)$$

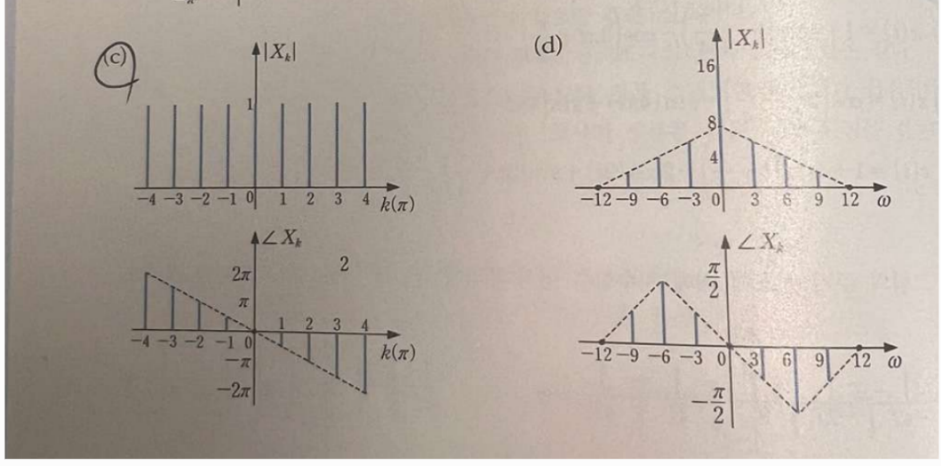
$n=2$

$$C_{+2}e^{j2\omega_0 t} + C_{-2}e^{-j2\omega_0 t} = 2|j|\cos(4\pi t + \arg j)$$

$$|j| = 1,\ \arg j = \frac{\pi}{2} \rightarrow 2\cos\left(4\pi t + \frac{\pi}{2}\right)$$

$$x(t) = 2 + 2\sqrt{2}\cos\left(2\pi t + \frac{\pi}{4}\right) + 2\cos\left(4\pi t + \frac{\pi}{2}\right)$$





$$|X_k| \begin{cases} 1 & k = \pm 1, \pm 3 \\ 2 & k = \pm 2 \\ 1 & k = 0 \end{cases}$$

$$\arg X_k = \begin{cases} +\frac{3}{4}\pi & k = -3 \\ -\frac{\pi}{6} & k = -2 \\ +\frac{\pi}{6} & k = -1 \\ +\pi & k = 0 \\ -\frac{\pi}{6} & k = 1 \\ \frac{\pi}{6} & k = 2 \\ -\frac{3}{4}\pi & k = 3 \end{cases}$$

$$X_k = |X_k| e^{j \angle X_k} = \begin{cases} e^{j\frac{3}{4}\pi} & k = -3 \\ 2e^{-j\frac{\pi}{6}} & k = -2 \\ e^{j\frac{\pi}{6}} & k = -1 \\ -1 & k = 0 \\ e^{-j\frac{\pi}{6}} & k = 1 \\ 2e^{j\frac{\pi}{6}} & k = 2 \\ e^{-j\frac{3}{4}\pi} & k = 3 \\ 0 & \text{otherwise} \end{cases}$$

$$X_0 = -1$$

$$x_{\pm 1} = e^{-j\frac{\pi}{6} + j\pi t} + e^{j\frac{\pi}{6} - j\pi t} = 2\cos\left(\pi t + \frac{\pi}{6}\right)$$

$$X_{\pm 2} = 2e^{j\frac{\pi}{6} + 2\pi t} + 2e^{-j\frac{\pi}{6} - j2\pi t} = 4\cos\left(2\pi t - \frac{\pi}{6}\right)$$

$$X_{\pm 3} = e^{-j\frac{3}{4}\pi + j3\pi t} + e^{j\frac{3}{4}\pi - j3\pi t} = 2\cos\left(3\pi t - \frac{3}{4}\pi\right)$$

$$x(t) = -1 + 2\cos\left(\pi t + \frac{\pi}{6}\right) + 4\cos\left(2\pi t - \frac{\pi}{6}\right) + 2\cos\left(3\pi t - \frac{3\pi}{4}\right)$$

(1)

$$|X_k| = 1, \quad \angle X_k = -\frac{\pi}{2}k$$

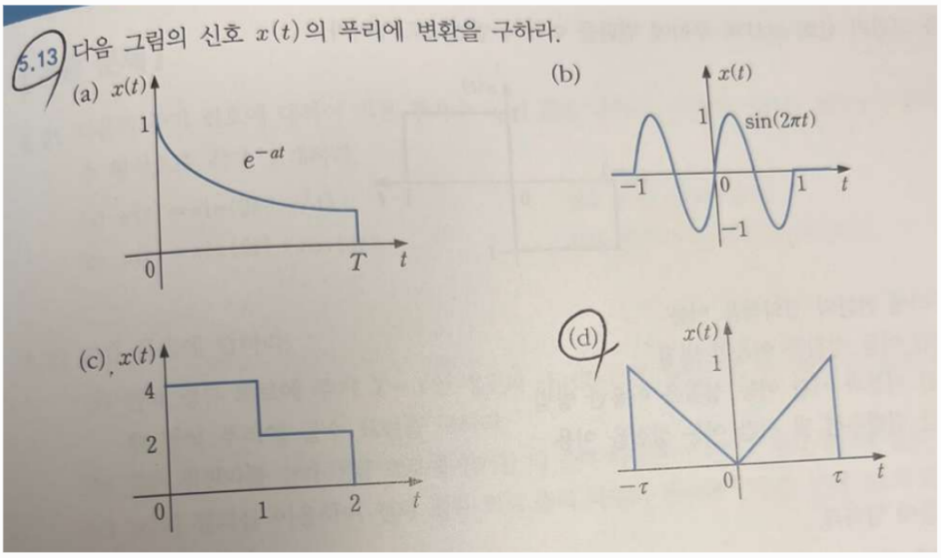
$$X_k = e^{-j(\pi/2)k}$$

$$x(t) = \sum_{k=-4}^{4} X_k e^{jk\omega_0 t} = X_0 + \sum_{k=1}^{4} \left[X_k e^{jk\omega_0 t} + X_{-k} e^{-jk\omega_0 t}\right]$$

$$X_k e^{jk\omega_0 t} + X_{-k} e^{-jk\omega_0 t} = 2\cos\left(k\pi t - \frac{\pi}{2}k\right)$$

$$x(t) = X_0 + 2\sum_{k=1}^{4} \cos\left(k\pi t - \frac{\pi}{2}k\right)$$

$$x(t) = 1 + 2\sin(\pi t) - 2\cos(2\pi t) - 2\sin(3\pi t) + 2\cos(4\pi t)$$



(2)

$$x(t) = \begin{cases} e^{-a} \cdot 0 \le t \le T \\ 0, \text{ otherwise} \end{cases}$$

$$X(w) = \int_0^T e^{-(a+jw)t}\,dt = \left[\frac{e^{-(a+jw)t}}{-(a+jw)}\right]_0^T$$

$$= \frac{1 - e^{-(a+jw)T}}{a+jw}$$

(d)

$$x(t) = \begin{cases} \frac{|t|}{\tau} & |t| \le \tau \\ 0, \text{ otherwise} \end{cases}$$

$$X(w) = \int_{-\infty}^{\infty} x(t)e^{-jwt}\,dt = 2\int_0^T \frac{t}{T}\cos(wt)\,dt$$

$$\int_0^T t\cos(wt)\,dt = \left[\frac{t\sin(wt)}{w}\right]_0^T - \int_0^T \frac{\sin(wt)}{w}\,dt$$

$$X(w) = \frac{2}{T}\left(\frac{T\sin(wT)}{w} + \frac{\cos(wT)-1}{w^2}\right) =$$

$$2\frac{\sin(wT)}{w} + 2\frac{\cos(wT)-1}{Tw^2}$$

$$\therefore X(w) = \frac{2(T\sin(wT)+\cos(wT)-1}{Tw^2}$$

5.18 다음 물음에 답하라.

(a) 푸리에 변환의 컨벌루션 성질을 이용해 다음 $X(\omega)$의 푸리에 역변환 $x(t)$를 구하라.

$$X(\omega) = \frac{1}{(a+j\omega)^2}$$

(b) 푸리에 변환의 미분 성질을 이용해 문제 (a)를 반복하라.

(c) 미분 성질을 이용해 다음 푸리에 변환의 역변환을 구하는 일반식을 구하라.

$$X(\omega) = \frac{1}{}$$

$\frac{1}{(a+j\omega)^m}$

(a)

$$X(\omega) = \frac{1}{(a+j\omega)^2}$$

$$x_1(t) = e^{-at}u(t) \longleftrightarrow X_1(\omega) = \frac{1}{a+j\omega}$$

$$x(t) = (x_1 * x_1)(t) = \int_{-\infty}^{\infty} x_1(\tau)\, x_1(t-\tau)\, d\tau$$

$$= \int_0^t e^{-a\tau} e^{-a(t-\tau)}\, d\tau = e^{-at}\int_0^t d\tau = te^{-at}$$

$$\therefore x(t) = te^{-at}u(t)$$

(b)

$$\mathcal{F}\{tx(t)\} = j\frac{d}{d\omega}X(\omega)$$

$$x_1(t) = e^{-at}u(t) \longleftrightarrow X_1(\omega) = \frac{1}{a+j\omega}$$

$$\mathcal{F}\{te^{-at}u(t)\} = j\frac{d}{d\omega}\left[\frac{1}{a+j\omega}\right] = j\left(-\frac{j}{(a+j\omega)^2}\right)$$

$$= \frac{1}{(a+j\omega)^2}$$

$$X(\omega) = \frac{1}{(a+j\omega)^2} \qquad x(t) = te^{-at}u(t)$$

(c)

$$\mathcal{F}\{t^n x(t)\} = j^n \frac{d^n}{d\omega^n}X(\omega)$$

$$x_1(t) = e^{-at}u(t) \longrightarrow X_1(\omega) = \frac{1}{a+j\omega}$$

$$\frac{d^{m-1}}{d\omega^{m-1}} \cdot \frac{1}{a+j\omega} = (-1)^{m-1}(m-1)!\,\frac{j^{m-1}}{(a+j\omega)^m}$$

X_1 를 $(m-1)$ 번 미분

$$\frac{d^{m-1}}{d\omega^{m-1}} \frac{1}{a+j\omega} = (-1)^{m-1}(m-1)! \frac{j^{m-1}}{(a+j\omega)^m}$$

$$\therefore \mathcal{F}\{t^{m-1}e^{-at}u(t)\} = j^{m-1}\frac{d^{m-1}}{d\omega^{m-1}}\left[\frac{1}{a+j\omega}\right]$$

$$= (m-1)!\frac{1}{(a+j\omega)^m}$$

$$\mathcal{F}^{-1}\left\{\frac{1}{(a+j\omega)^m}\right\} = \frac{t^{m-1}}{(m-1)!}e^{-at}u(t)$$